

GenMol UDLM engineering exploration

engineering-v9-ct-ce-objectives | complete | 2026-09-07T11:25:39.800704+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

CT and clean CE share the training settings recorded in this protocol. Equal updates and examples do not imply equal compute or independently optimized objectives; training losses have different scales and targets.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED ≥ 0.6 and SA ≤ 4 . Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Predeclared contrasts subtract CT from CE within the same seed and decoding branch. Every declared contrast is retained, including negative outcomes. Paired mean and sample SD are withheld unless all declared seed pairs define that metric; unavailable pairs remain visible. Sample SD describes seed spread, not a confidence interval. Pairing seed labels does not imply coupled molecular trajectories.

Prospective sampling design

`{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "objective_comparison": {"contrast": "CE minus CT for each seed, then equal-seed mean; report both contrasts even when negative."}, "primary": {"control_config": "ct_t100", "temperature": 1.0, "treatment_config": "ce_t100"}, "secondary": {"control_config": "ct_t050", "temperature": 0.5, "treatment_config": "ce_t050"}, "selection_history": "V5/V6 outcomes and V7 throughput were known before design; temperature 0.5 is an informed secondary engineering setting."}, "parameterization_controls": "Convert CE clean-denoiser logits to LOO at the current noisy state/time before temperature and top-p; CT logits already parameterize raw LOO.", "predictor_transitions_per_molecule": 128, "randomness": 0.0, "raw_loo_top_p": 1.0}`

The original V8 CT controller and campaign remain failed. This comparison uses its separately audited checkpoint plus a new completed V8b CE run; never label V8 campaign success.

Training source revisions differ through controller/reporting changes; model/training implementation and resolved common training settings are matched by recorded hashes.

128000 training exposures are configured accounting, not distinct molecules or an independent live row trace. Finite final tensors do not establish finiteness of every intermediate update.

V5/V6 informed this engineering design, especially temperature 0.5; this is not independent confirmation or a superiority claim.

Two seeds and 100 requests per setting are small; pairing is by seed/run, not molecule-level trajectories or independence of generated observations.

Report validity, uniqueness, quality and diversity under both strict SAFE decoding and released repair/largest-component selection. Quality counts unique QED $\geq$ 0.6 and SA $\leq$ 4 molecules divided by requests.

Equal updates/examples and shared hyperparameters do not imply equal compute or independently optimized CT/CE objectives. The baseline uses 50k MDLM updates and three 1000-request seeds.

Final UDLM seeds 0/1/2 remain reserved. No automatic generation follows this materialization, and no model promotion or longer training follows the results.

Objective comparison

CT / raw-LOO predicts clean leave-one-out probabilities. Clean CE predicts clean denoiser probabilities, converted to LOO logits at the current noisy state and time before temperature or top-p. The conversion adds no backbone evaluation. Shared sampling settings do not imply independently optimized objectives. Objective labels and checkpoint evidence below describe completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b"}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}, "CT": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_implementation_hashes_sha256": "c9641b7e447d1f4617599e4df338fde0010fe68f7877566de37d2e6c5a6ccd2b", "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdln_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
ct_t100	1600	128	128	0
ct_t100	1601	128	128	0
ce_t100	1600	128	128	0
ce_t100	1601	128	128	0
ct_t050	1600	128	128	0
ct_t050	1601	128	128	0
ce_t050	1600	128	128	0
ce_t050	1601	128	128	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
ct_t100	1600	CT / raw-LOO	raw_loo	ema
ct_t100	1601	CT / raw-LOO	raw_loo	ema
ce_t100	1600	clean CE	x0_denoiser	ema
ce_t100	1601	clean CE	x0_denoiser	ema
ct_t050	1600	CT / raw-LOO	raw_loo	ema
ct_t050	1601	CT / raw-LOO	raw_loo	ema
ce_t050	1600	clean CE	x0_denoiser	ema

Config	Seed	Objective	Parameterization	Weights
ce_t050	1601	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t100	2/2	0.9750 +/- 0.0212	0.9948 +/- 0.0074	0.4650 +/- 0.0636	0.9090 +/- 0.0001
ce_t100	2/2	0.9800 +/- 0.0141	0.9948 +/- 0.0073	0.4100 +/- 0.0424	0.9025 +/- 0.0036
ct_t050	2/2	0.9850 +/- 0.0071	0.9949 +/- 0.0071	0.5450 +/- 0.0212	0.8838 +/- 0.0028
ce_t050	2/2	0.9900 +/- 0.0000	0.9949 +/- 0.0071	0.5250 +/- 0.0495	0.8880 +/- 0.0059

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t100	2/2	0.3900 +/- 0.0566	1.0000 +/- 0.0000	0.2100 +/- 0.0141	0.8864 +/- 0.0068
ce_t100	2/2	0.6450 +/- 0.0778	1.0000 +/- 0.0000	0.2600 +/- 0.0707	0.8867 +/- 0.0049
ct_t050	2/2	0.7700 +/- 0.0141	1.0000 +/- 0.0000	0.4400 +/- 0.0141	0.8683 +/- 0.0082
ce_t050	2/2	0.6700 +/- 0.0566	1.0000 +/- 0.0000	0.3950 +/- 0.0354	0.8700 +/- 0.0007

Predeclared paired CE minus CT contrasts

Differences use the same seed and decoding branch, with equal seed weights. Values are on the metric's 0-to-1 scale: 0.01 is one percentage point for quality. All declared pairs must define a metric before its mean and sample SD are shown. SD is seed spread, not a confidence interval. Seed labels do not imply paired molecules.

primary | temperature 1.0: ce_t100 minus ct_t100

released_comparable

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.990000	+0.990000	+0.000000	completed

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1601	+0.960000	+0.970000	+0.010000	completed
uniqueness	1600	+1.000000	+1.000000	+0.000000	completed
uniqueness	1601	+0.989583	+0.989691	+0.000107	completed
quality	1600	+0.510000	+0.440000	-0.070000	completed
quality	1601	+0.420000	+0.380000	-0.040000	completed
diversity	1600	+0.909109	+0.900010	-0.009099	completed
diversity	1601	+0.908897	+0.905074	-0.003823	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	+0.005000	0.007071
uniqueness	2/2	+0.000054	0.000076
quality	2/2	-0.055000	0.021213
diversity	2/2	-0.006461	0.003731

strict

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.430000	+0.700000	+0.270000	completed
validity	1601	+0.350000	+0.590000	+0.240000	completed
uniqueness	1600	+1.000000	+1.000000	+0.000000	completed
uniqueness	1601	+1.000000	+1.000000	+0.000000	completed
quality	1600	+0.220000	+0.310000	+0.090000	completed
quality	1601	+0.200000	+0.210000	+0.010000	completed
diversity	1600	+0.891188	+0.883229	-0.007959	completed
diversity	1601	+0.881626	+0.890105	+0.008479	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	+0.255000	0.021213
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.050000	0.056569
diversity	2/2	+0.000260	0.011624

secondary | temperature 0.5: ce_t050 minus ct_t050

released_comparable

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.990000	+0.990000	+0.000000	completed
validity	1601	+0.980000	+0.990000	+0.010000	completed
uniqueness	1600	+0.989899	+1.000000	+0.010101	completed
uniqueness	1601	+1.000000	+0.989899	-0.010101	completed
quality	1600	+0.530000	+0.490000	-0.040000	completed
quality	1601	+0.560000	+0.560000	+0.000000	completed
diversity	1600	+0.885833	+0.892143	+0.006309	completed
diversity	1601	+0.881839	+0.883780	+0.001941	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	+0.005000	0.007071
uniqueness	2/2	+0.000000	0.014285
quality	2/2	-0.020000	0.028284
diversity	2/2	+0.004125	0.003089

strict

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.780000	+0.630000	-0.150000	completed
validity	1601	+0.760000	+0.710000	-0.050000	completed
uniqueness	1600	+1.000000	+1.000000	+0.000000	completed
uniqueness	1601	+1.000000	+1.000000	+0.000000	completed
quality	1600	+0.450000	+0.370000	-0.080000	completed
quality	1601	+0.430000	+0.420000	-0.010000	completed
diversity	1600	+0.874062	+0.869567	-0.004494	completed
diversity	1601	+0.862439	+0.870508	+0.008069	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	-0.100000	0.070711
uniqueness	2/2	+0.000000	0.000000
quality	2/2	-0.045000	0.049497
diversity	2/2	+0.001787	0.008884

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_t100	1600	completed	100	100	0.5100	0.2200	12.24
ct_t100	1601	completed	100	100	0.4200	0.2000	13.86
ce_t100	1600	completed	100	100	0.4400	0.3100	12.40
ce_t100	1601	completed	100	100	0.3800	0.2100	11.93
ct_t050	1600	completed	100	100	0.5300	0.4500	14.36
ct_t050	1601	completed	100	100	0.5600	0.4300	13.98
ce_t050	1600	completed	100	100	0.4900	0.3700	12.52
ce_t050	1601	completed	100	100	0.5600	0.4200	13.92

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: 718893198b0e08a2405e33177c64989499914207ff39b6880edbc5eff2b35c7d

v9-ct-t100 / seed 1600: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t100/seed_1600/raw_samples.csv", "sha256": "f8b3994c1a65b0070cf248e44d795a4071ec06f521a7b693b8aba10d1457445d", "size_bytes": 30618}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ct-t100 / seed 1601: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay":

0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 2}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t100/seed_1601/raw_samples.csv", "sha256": "13a78c24b80be0063ee9e44b9aace057d322c35acb2af114efef1f50cb374ee6", "size_bytes": 29880}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ce-t100 / seed 1600: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 7}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t100/seed_1600/raw_samples.csv", "sha256": "7680236fc4efc888b85c91569430d40ad0e44d9337983249957f9fd72a8b27c8", "size_bytes": 32879}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v9-ce-t100 / seed 1601: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256":

"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1256}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t100/seed_1601/raw_samples.csv", "sha256": "91de05669cdc5be79830e1c6173a395309ed478941d3e93cbd775556d52c84b3", "size_bytes": 32925}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v9-ct-t050 / seed 1600: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t050/seed_1600/raw_samples.csv", "sha256": "11dbab23177d6591de03c5097edb9d7f4f7a000554f19906f07c043261cec716", "size_bytes": 32877}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ct-t050 / seed 1601: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python",

"used_memory_mib": 1256}}, "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, {"launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpu", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, {"logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, {"physical_index": 6}], {"inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, {"objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t050/seed_1601/raw_samples.csv", "sha256": "13d0d3fb32634f20490389e74883f8ba88a196bb8cf565bdca0b5c8f0d9c028e", "size_bytes": 33609, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}}

v9-ce-t050 / seed 1600: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, {"denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, {"model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, {"gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, {"launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpu", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, {"logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, {"physical_index": 7}], {"inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, {"objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t050/seed_1600/raw_samples.csv", "sha256": "32b342171299c7bfd745c16007ade327ffdb67b94c14047749f140898356c205", "size_bytes": 32079, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}}

v9-ce-t050 / seed 1601: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, {"denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, {"model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, {"gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1256}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, {"launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpu",

```
"max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison":  
"strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index":  
"6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE",  
"parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t050/seed_1601/raw_samples.csv", "sha256":  
"41af77517633bd59388930f20f0435a3b57d86cb354a2e1b9ec5095a7f4e98dd", "size_bytes": 33179}, "sampling_budget": {"corrector_steps_per_molecule": 0,  
"gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata":  
{"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser",  
"schema_version": 1}}
```